SCHEME OF B.Voc. MULTIMEDIA

(GRAPHICS & ANIMATION)

(SEMESTER SYSTEM)

Semester I								
Paper Code	**Title**	**Generic/ Skill Component**	**Theory/ Practical**	**Internal (Theory)**	**External (Theory)**	**Internal (Practical)**	**External (Practical)**	**Credit**
*GEN -101	Communication Skills	Generic	Theory	20	80	--	--	6
*GEN -102	Fundamentals of Information Technology	Generic	Theory	20	80	--	--	6
MGA - 101	Understanding of Multimedia & Animation	Skill	Theory	20	80	--	--	6
MGA - 102	CorelDraw and Photoshop	Skill	Theory	20	80	--	--	6
MGA - 103	Practical based on Paper MGA -101 & MGA- 102	Skill	Practical	--	--	20	80	6
SEMESTER II								
*GEN 201	Soft Skills and Personality Development	Generic	Theory	20	80	--	--	6
GFC 104	Programming in C	Generic	Theory	20	80	--	--	6
MGA-104	Cinematography	Skill	Theory & Practical	10	40	10	40	6
MGA-105	Video and Audio Editing	Skill	Theory & Practical	10	40	10	40	6
MGA-106	Art Foundation : Drawing	Skill	Theory & Practical	10	40	10	40	6
	Summer Training [OJT] After 2nd Semester Examinations							NC**

***Refer to Generic Components Common to all B.Voc. Courses**

**** Non Credit**

Job Role: Web Designer, 2D Animator, 3d Modeler & Architectural Designer.

Semester III								
Paper Code	**Title**	**Generic/ Skill Component**	**Theory/ Practical**	**Internal (Theory)**	**External (Theory)**	**Internal (Practical)**	**External (Practical)**	**Credit**
*GEN 301	Value Education And Human Rights	Generic	Theory	20	80	--	--	6
GFC-202	Internet Technology & Web Designing	Generic	Theory	20	80	--	--	6
MGA - 201	Flash	Skill	Theory	20	80	--	--	6
MGA - 202	Dreamweaver	Skill	Theory	20	80	--	--	6
MGA - 203	Practical Based on MGA-201 & MGA-202	Skill	Practical	--	--	20	80	6
SEMESTER IV								
*GEN 401	Environmental Studies	Generic	Theory	20	80	--	--	6
GFC-204	Animation Theory	Generic	Theory	20	80	--	--	6
MGA 204	3ds Max: Modeling	Skill	Theory & Practical	10	40	10	40	6
MGA 205	3ds Max: Texturing, Lighting & Rendering	Skill	Theory & Practical	10	40	10	40	6
MGA 206	Practical Based on MGA-204 & MGA-205	Skill	Theory & Practical	10	40	10	40	6
	Summer Training [OJT] After 2nd Semester Examinations							NC**

***Refer to Generic Components Common to all B.Voc. Courses**

**** Non Credit**

Job Role: Java Developer, 3D Animator, VFX Editor.

Semester V								
Paper Code	**Title**	**Generic/ Skill Component**	**Theory/ Practical**	**Internal (Theory)**	**External (Theory)**	**Internal (Practical)**	**External (Practical)**	**Credit**
*GEN 501	Critical Thinking And Elementary Statistics	Generic	Theory	20	80	--	--	6
GFC-302	Internet Programming With Java	Generic	Theory	20	80	--	--	6
MGA-301	3d Max Animation (Practical)	Skill	Theory	20	80	--	--	6
MGA-302	Visual Effects (Practical)	Skill	Theory	20	80	--	--	6
MGA-303	Pre-Production(Theory)	Skill	Practical	--	--	20	80	6
SEMESTER VI								
*GEN 601	Entrepreneurship Development Programme	Generic	Theory	20	80	--	--	6
MGA-304	Project & Internship.	Skill	Theory	20	80	--	--	6

***Refer to Generic Components Common to all B.Voc. Courses**

**** Non Credit**

Job Role: Java Developer, 3D Animator, VFX Editor.

Understanding of Multimedia & Animation

Paper Code: MGA-101

Course Objective: To familiarize the students with various approaches, methods and techniques of Animation Technology. Mastering traditional & digital tools to produce stills and moving images. To apply Audio and Video Production Techniques to an Animation Project.

Instructions:

- The syllabus of this paper has been divided into FOUR units.
- Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
- The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
- All questions carry equal marks.

UNIT I

Introduction to Multimedia: Multimedia and Hypermedia, Overview of Multimedia Software Tools. Authoring Tools-Multimedia Authoring, Editing and Authoring Tools.

Graphics and Image Data Representations: Graphics/Image Data Types, File Formats-GIF, JPEG, PNG, TIFF, EXIF, PDF, WMF, BMP.

Color in Image and Video: Color Science, Color Models in Images-RGB, CMY, CMYK, Color Models in Video.

UNIT II

Fundamental Concepts in Video: Types of Video Signals, Analog Video-NTSC, PAL, SECAM, Digital Video.

Basics of Digital Audio: Digitization of Sound, MIDI, Quantization and Transmission of Audio. Multimedia Data Compression: Lossless Compression-Run Length Encoding, Variable Length Coding, Dictionary Based Coding, Arithmetic Coding. Lossy Compression Algorithms-Distortion Measures, Rate Distortion Theory, Quantization, Transform Coding, Wavelet Coding.

UNIT III

Multimedia Communication & Retrieval: Computer and Multimedia Networks-Basics of Computer and Multimedia Networks, Multiplexing Technologies, LAN & WAN, Access Networks, Common Peripherals Interfaces.

Wireless Networks: Analog Wireless Networks, Digital Wireless Networks, TDMA GSM, CDMA, 3G, Wireless LAN. Radio Propagation Models, Multimedia Over Wireless Networks.

UNIT IV

History of Animation, Types of Animation, Skills for Animation Artist, Basic Principles of Animation, Animator's Drawing Tools, Rapid Sketching & Drawing, Developing Animation Character, Essentials & Qualities of Good Animation Characters, Three Dimensional Drawings of Characters.

References:

1. Ze-Nian li, Mark S. Drew,2004: *Fundamentals of Multimedia*, PHI.
2. Chris Patmore , 2009, *The Complete Animation Course* , Barons Educational Series
3. Vaughan, Tay, 2008 : *Multimedia Making it Work*, Osborne Publishers.
4. Judith Jeffcoate, 2007 : *Multimedia in Practice, Technology and Applications*, PHI.

CorelDraw and Photoshop

Paper Code: MGA-102

Course Objective: CorelDraw & Photoshop students are expected to use their own designing skills together with this application to create stunning illustrations, logos, advertisements, etc. Also students & professionals will be using advanced tools for creating drawings, advanced techniques for text manipulation, advanced output options for printing etc.

Instructions:

- The syllabus of this paper has been divided into FOUR units.
- Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
- The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
- All questions carry equal marks.

UNIT-I

Introduction to CorelDraw: CorelDraw Screen, Drawing Lines, Selecting Objects, Creating Artistic Text, Formatting Text, Working with Shapes, Working with Rectangles, Creating Ellipses and Circles, Drawing Polygons, Drawing Spirals, Editing and Transforming Shapes, Saving Files, Printing Files.

Controlling CorelDraw Environment: Viewing and Moving Toolbars, Working with Docker Window, Setting up Page Layout, Defining Outlines.

UNIT-II

Working with Shapes and Curves: Drawing and Editing Freehand Curves, Editing Shapes, Editing Curves, Working with Bitmap Images in CorelDraw, Applying Effects to Bitmaps.

PHOTO-PAINT and 3D Basics: Creating a New Bitmap Image, Setting Colors and Fills, Painting Text , Painting Shapes, General 3D Concepts.

UNIT-III

Adobe Photoshop Basics: Introducing Adobe Photoshop, Exploring the Photoshop Workspace, Touring the Workspace, Exploring the Photoshop Menu Bar.

Image Basics: Exploring File Types, Understanding Image Files, Understanding Video Files, Creating and Opening Images, Resizing Files and Adjusting Resolution, Understanding Colors Basics, Adjusting with Histograms, Working in Different Color Modes.

UNIT-IV

Creating Selections: Using the Selection Tools, Working with Lasso Tools and Quick Selection Tools, Refining and Adjusting Selection.

Applying Sharpness and Blur Adjustment: Using Sharpening Filter, Using Blur Filter, Introducing the Blur Gallery, Using the Healing Brush Tools

References :

1. Dave Karlins,1999:*Sam'sTeach Yourself CorelDraw9 in 24 hours,* Sams Publishing.
2. Brad Dayley & DaNae Dayley,2012:*Adobe Photoshop CS6 BIBLE,* Wiley.
3. Altman, 1999 : *Mastering Corel-Draw,* BPB.
4. Gary David Bouton, 2012: *CorelDRAW X6 The Official Guide*, McGraw Hill Professional.
5. Romaniello, 2002 : *Photoshop 7 – Savvy*, Sybex.
6. Adobe Creative Team, 2012: *Adobe Photoshop CS6 Classroom in a Book*, Adobe Press.

Practical based on Paper MGA-101 & MGA-102

Paper Code: MGA-103

There will be a practical examination in this paper along with semester examination. External examiner will evaluate the skills of the students in the field of CorelDraw and Photoshop. Examiner will give on-the-spot assignments to the students. Besides this a viva voce (oral examination) of each student one by one will also be taken by the examiner for testing the knowledge of the students.

Semester II

Paper Title: Programming in C

Paper Code: GEN-104

Objective: To expose the students to the concept of C language programmer. To fulfill the need and importance of creating an awareness of these resources and to maximize the same to enable the students meet the challenges of the modern world.

Instructions:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

UNIT I

Fundamentals of 'C': I/O statements, assignment statements, Constants, variables and data types, Operators and Expressions, Standards and Formatted statements, Key word, Data Types and Identifiers.

UNIT II

Control Structures: Introduction, Decision making with IF – statement, IF – Else and Nested IF, While and do-while, for loop, Jump statements, break, continue, goto, Switch Statement. Functions: Introduction to functions, Global and Local Variables, Function Declaration, Standard functions..

UNIT III

Preprocessor Directives: Introduction and Use, Macros, Conditional Preprocessors, Header Files, Arrays: Introduction to Arrays, Array Declaration, Single and Multidimensional Array, Arrays of characters Pointers: Introduction to Pointers, Address operator and pointers, Declaring and Initializing pointers.

UNIT IV

Structures and Unions: Declaration of structures, Accessing structure members, Structure Initialization, Arrays of structure, nested structures, structure with pointers, Unions.
Strings: Introduction, Declaring and Initializing string variables, Reading and writing strings, String handling functions.

References:

1. Byron S. Gottfried, 1996 : Programming in C, McGraw Hills Publishers, New York.
2. Salaria, R. S. : Test Your Skills in C, Khanna Book Publishing Co. (P.) Ltd.,New Delhi.
3. Kanetkar Yashwant, 2010 : Let us Exploring C, BPB Publications, New Delhi
4. Balaguruswami C., 2008 : Programming with C Language, Tata McGraw Hill, New Delhi.
   Somashekara, M.T. , 2008 : Programming in C, Prentice Hall of India.

Cinematography
Paper Code: MGA-104

Course Object: The subject also covers basic camera terms, Light and visual storytelling techniques, including the use of the video technology camera and digital video editing equipment and software.

Instructions:

- The syllabus of this paper has been divided into FOUR units.
- Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
- The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
- All questions carry equal marks.

UNIT I

Writing with Motion: Conceptual Tools of Cinematography, Frame, Lens, Light and Color, Texture, Movement, Establishment, Point-of-View.

Shooting Methods: Cinematic, The Frame, Cinema as a Language, Shooting Methods.

Visual Language: Design Principles, Forces of Visual Organization, Rules of Composition.

Language of the Lens: Lens and the Frame, Image Control at the Lens.

UNIT II

Visual Storytelling: Visual Metaphore, Lighting as Storytelling, Lighting as Visual Metaphore.

Cinematic Continuity: Shooting for Editing, Prime Directive, Screen Direction, Turnaround, Planning Coverage, Issues in Continuity and Shooting, Types of Cuts.

Lighting Basics: Fundamentals of Lighting, Lighting Terminology, Aspects of Light, Basic Lighting Techniques, Motivated Light, Day Exteriors, Lighting for High Def. Video.

Lighting Sources: Tools of Lighting, Daylight Source, Xenons, PARs, Soft Lights, Color Correction Fluorescents.

UNIT III

HD Cinematography: High Def. and Standard Def. Videos, Analog & Digital Video, Types of Video Sensors, Digital Video, Shooting Formats, Digital Compression, Monitoring on the Set, Video Signal, Setting Up a Color Monitor, Camera with Balance, Timecode & Edgecode, Compression & Codecs.

Exposure: Light as Energy, Exposure, ISO, and Lighting Relationships, Light & Film, Densitometry, Brightness Perception, Contrast, Zone System, Gray Scale, Exposure & Camera.

Camera Movement: Basic Techniques, Moves, Moving Shots, Camera Mounting, Crab Dolly, Cranes, Car Shots, Ariel Shots, Motion Control.

UNIT IV

Image Control: Color Printing, Controlling Color & Contrast, LookUp Tables, Camera Filter Types, Color Temperature & Filtration, Image Control with Camera.

Optics & Focus: Basic of Optics, Focus, Depth-of-Field, Macrophotography.

Set Operations: Director of Photography, Team, Time Code Slates.

Technical Issues: Flicker, Filming Practical Monitors, Shooting Process, Dimmers, High Speed Photography, Underwater Filming, Image Quality, Effects, Sun Location with Compass, Transferring Film to Video**,** Film Formats.

References:

1. Blain Brown, 2013: *Cinematography, Theory And Practice*, Focal Press.
2. Kris Malkiewicz, M. David Mullen, 2009: *Cinematography: Third Edition*, Simon and Schuster.
3. Joseph v. Mascelli, 1998: *The Five C's of Cinematography,* Silman-James Press.
4. Viktor Shklovkii, 2008: *Literature and Cinematography.* Dalkley Archive Press.

Video and Audio Editing

Paper Code: MGA-105

Course Objectives:-

- Prepare for and make a movie using good movie making techniques
- Demonstrate understanding of the development process for making movies
- Use a video camera to capture video
- Upload digital video to a computer
- Apply knowledge of Adobe premier as a video editing and movie making software
- Apply knowledge of Sound Forge to add and edit audio
- Export a movie and burn to a disc.

Instructions:

- The syllabus of this paper has been divided into FOUR units.
- Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
- The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
- All questions carry equal marks.

UNIT I

Introduction to Video Editing, Non-Linear Editing: Meaning and Process, Hardware Requirements for NLE, Connecting Editing Equipment, Creating Time Line, Cut, Fade, Mix and Wipe, Introduction to Editing Aesthetics.

Unit II

Main Tools of Editing - Selection, Range Select, Rolling, Edit, Razor, Hand Tool, Cross Fade, In Point, Zoom Tools , Video Transitions, Titling and Graphics, Rendering and Authoring, Giving Effects, Mixing Video, Editing on The Tape, Hard Disk.

Unit III

Nature of Digital Sound, Meaning of Mono, Stereo and Surround Sound, Various Formats of Digital Sound, Editing Tools: Cut, Cross Fade, Mix, Zero-Crossing Edits, Multiple Track Editing Process of Equalization, Applying Effects to Sound.

Unit IV

Introduction to Audio Console, Audio Mixing, Deleting Unwanted Audio, Getting Output, Fades and Cross-Fades, Giving Audio Effects, Editing Sound Using Tape, Hard Disk, Introduction to Sound Aesthetics, Introduction to Sound Forge, Nuendo

References:

1. Jhon Rice and Brian Mckernan, 2003: *Editing Digital Video,* McGraw-Hill Companies
2. Ken Dancyger, 2011:*The Technique of Film and Video Editing: History, Theory, and Practice*, Focal Press .
3. Simon Langford, 2013:*Digital Audio Editing*, Focal Press.
4. Sam McGuire, David Liban,2010:*The Video Editor's Guide to Soundtrack Pro: Workflow, Tools and Techniques*, Focal Press.
5. Dave D. Peck, 2012: *Video Editing with Adobe Premiere*, Cengage Learning.
6. Scott R. Garrigus, 2010: *Sound Forge Power*, Firewall Media.

Art Foundation: Drawing

Paper Code: MGA-106

Course Objective:- This course is an introduction to basic drawing materials, practices, and concepts. The class will focus on the process of perception in particular as a means to achieve a successful drawing. Various techniques of using graphite drawing and the use of line, value shading, and composition will be constantly emphasized and practiced, using assorted classroom and outdoor arrangements as a basis for study.

UNIT I

Free Hand Sketching From Real Objects: Building, Vehicles, Chair, Table, Trees etc.
Preparing Colour Charts.
Kind of Design- 2 D & 3 D Design

UNIT II

Designing Letter Heads and Visting Cards, Designing Book Covers and Folders.

UNIT III

Landscape on the Spot & Landscape Form Memory (Hills, Trees, Hut, Rivers etc.), Layout Designing of Newspapers, Magazines, Print Advertisements.

UNIT IV

Poster Making on Various Themes and Designing of Hoarding Making.

Note:- There will be a practical examination in this paper along with semester examination. External examiner will evaluate the skills of the students in the field of drawing. Examiner will give on-the-spot assignments/tasks to the students. Besides this a viva voce (oral examination) of each student one by one will also be taken by the examiner for testing the knowledge of the students.

Semester III

Internet Technology & Web Designing

Paper Code: GFC-202

Course Object: Understand scripting languages that are commonly used on the World Wide Web. Design, construct and maintain interactive web pages and web sites. Acquire skills and knowledge of design principles to support the building, testing and refining of Web pages.

UNIT-I

Introduction to Internet: Internet Overview, Evolution & History of Internet, Growth of Internet, Internet Services, Anatomy of Internet, Internet Applications, Impact of Internet.

Internet Technology and Protocol: OSI Reference Model, TCP/IP Reference Model, Data Transmission, Switching, Routers and Gateways, Network Protocols, Internet Addressing.

Internet Connectivity: Different Types of Connections, Hardware and Software requirements of Internet, Levels of Internet Connectivity, Internet Service provider.

UNIT-II

Computer Network: Network Definition, Common Terminology- LAN, WAN, Network Components, Network Topologies, Types of Network Architecture, Different Medium of Communication, Network Security.

WWW and Web browser: WWW, Evolution of Web, Basic Elements of WWW, Web Browsers and Search Engines.Electronic Mail: Email Basics, Concepts of Email Server, Structure of Email Message, Working of Email System, Email Security.

UNIT-III

Web Publishing and Browsing: Overview, SGML, Web Hosting, URL, CGL, Components of web Publishing, Web Page Design consideration and Principles.

HTML Programming Basics: HTML Page Structure, HTML Lists, HTML Tables, HTML Links, HTML Forms, HTML Frames.

UNIT-IV

Internet Tools and Multimedia: Current Trends on Internet, Interactivity Tools, Multimedia and Animation.

Remote Login: introduction to Remote login, Introduction to Telnet, File Transfer Protocol.

Internet and Web Security: Overview of Internet Security, Email Threats, Web Security and Privacy Concepts, Firewall, Cryptography, Digital Signatures, Authentication and Access Control, Virus, Internet Security and Management Concepts.

References:

1. ISRD Group, 2011: *Internet Technology and Web Design*, Tata McGraw-Hill.
2. Tom Craig, 2003: *Internet: Technology, People, Process*, Black Rabbit Books.
3. Valerie Bodden, 2008: *Internet,* The Creative Company.
4. Ewaryst Tkacz, Adrian Kapczynski, 2009: *Internet-Technical Development and Applications,* Springers Publications.

Flash

Paper Code: MGA-201

Course Objective: This subject, delegates will be able to develop efficient, compact Flash movies; Create, import, and work with graphics; Import graphics from other applications; Work with text; Use layers and frames; Use masks; Create and use symbols, libraries, and instances; Build frame by-frame animations; Motion tween; Shape tween; Work with motion guides; Create navigation buttons; Work with simple ActionScript commands; Use components; Add sounds to documents; Import Video; Publish a Flash document.

UNIT-I

Understanding the Adobe Flash: The Topography of Flash CS5, Bitmap Handlers, Vector Based Drawing Programs, Flash in Internet Evolution, Exploring Companion Technologies, Planning Flash Projects.

Mastering in Flash Environment: Interface Fundamentals, Managing Windows and Panels, The Tools Panel, Document and Timeline Window, Drawing in Flash, Using Shapes and Drawing Tools, Using Fill and Stroke Controls, Understanding Symbols Instances and Library.

UNIT –II

Applying Color and Graphics: Introducing Color basics, Working in the Swatches Panel,Using the Color Panel, Modifying Graphics, Transforming Gradients and Bitmap Fills, Manipulating Symbols in 3D Space.

Integrating Media Files with Flash: Adding Sound, Importing Sound in Flash, Synchronizing Audio to Animation, Stopping Sounds, Editing Audio in Flash.

UNIT-III

Defining Vector, Bitmaps and Video: Importing and Copying Bitmaps, Setting Bitmap Properties, Bitmap Compression, Converting Raster to Vector.

Displaying Video: Integrating Video, Importing the Video, Compressing Video with Adobe Media Encoder CS5.

Adding Basic Interactivity to Flash Movie: Understanding Action and Event Handlers, First Five Actions, Building Timelines and Interactions.

UNIT-IV

Distributing Flash Movies: Testing Flash Movies, Publishing your Flash Movies, Selecting Formats, Integrating Flash Content with Web Pages, Writing Markup for Flash Movies, Detecting the Flash Player. Using The Flash Player: Using the Stand-Alone Flash Player, Using the Flash Player Plug In for Web Browsers, Using the Flash Player Setting.

References:

1. Todd Perkins, 2011: *Adobe Flash Professional CS5, Bible*, Wiley India.
2. Adobe Creative Team, 2013: *Adobe Flash Professional CS6-Classroom in a Book,* Pearson.
3. AGI Creative Team, 2012: *Adobe Flash Professional CS6 Digital Classroom,* Wiley India.
4. Alex Micheal, 2013: *Animating with Flash8:Creative Animation Techniques,* Focal Press.

Dreamweaver

Paper Code: MGA-202

Course Objectives:

- Have hands-on knowledge in developing simple and comprehensive Internet Web sites.
- Be able to plan, design, and develop Web sites.
- Acquire creative skills in design, layout, and interactivity of Web pages.
- Integrate varied software applications such as Web editor, graphic, and multimedia software.
- Create a portfolio of Web sites that demonstrate competency in the software application.
- Understand fundamental trends of technological evolution of Web development.

UNIT-I

Introduction: Explore the Dreamweaver Workspace, Features of Dreamweaver, Website Development Process, Adding a Folder to the Website, Creating the Homepage, Adding Pages to a Website, Developing a Webpage, Creating Head Content and Set Page Properties, Create, Import and Format Text, Adding Links to Webpages, Using the History Panel, Editing the Code. Adding Image, Enhancing the Image and Use Alternate Text, Creating a Website Color Palette.

UNIT-II

Cascading Style Sheets: Introduction, Types, CSS Styles, Panel, Advantages & Disadvantages of CSS, Adding Rules, Creating Pages Using CSS Layouts.
Links: External& Internal Links, Create, Modify and Copy Spry Menu Bar, Create an Image Map, Tables, Creating a Table, Properties, Merging Cells, Insert &Align Images in a Table, Insert Text & Format Cell Content.

UNIT-III

Forms: Plan and Create Forms, Edit & Format the Forms, Form Objects, Test & Process Forms.
Positioning Objects with AP Div:Insert an AP Div, Position and Size, Adding Content to an AP Div, Adding Images, AP Div Properties, Using AP Elements Panel, Cloak Files, Check Files Out and In.

UNIT-IV

Media and Interactivity:Adding &Modifying Flash Objects, Inserting Rollover Images, Behaviors Panel, Adding Video and Sounds, Creating Pages Using Templates, Editable Regions, Optional Regions.

References:

1. Sherry Bishop, 2012: *Adobe Dreamweaver CS6 Reveald*, Cengage Learning.
2. Adobe Creative Team, James J. Maivald, 2012: *Adobe Dreamweaver CS6 Classroom in a Book*,
3. Adobe Press.
4. Jeremy Osborn, AGI Creative Team, 2012: *Adobe Dreamweaver CS6 Digital Classroom*,
5. Vickie Ellen Wolper, 2011: *Artistic Web Design Using Adobe Dreamweaver and Photoshop,*
6. Jones & Brelett Learning.
7. Patricia Robertson, 2014: *Dreamweavers*, Dreamweaver Press.

Practical Based on MGA-201 & MGA-202

Paper Code: MGA-203

There will be a practical examination in this paper along with semester examination. External examiner will evaluate the skills of the students in the Subject of Adobe Flash and Dreamweaver. Examiner will give on-the-spot assignments to the students. Besides this a viva voce (oral examination) of each student one by one will also be taken by the examiner for testing the knowledge of the students.

Semester IV

Animation Theory

Paper Code: GFC-204

Course Objective: To understand the complex, technical and aesthetic components of the design of animation.

UNIT-I

Development: Idea Creation, Evolving a Storyline, Scriptwriting.

Character Design: Evolution of 2D and Evolution of 3D Character design, Animation Style, Concept and Environment Design.

Project Financing: Animation Markets, Scheduling and Budgeting, Investment, Marketing and Distribution Possibilities. Short and Independent Film Developing.

UNIT-II

Rules of Filmmaking: Camera Positions, Camera Lenses, Lighting and Filters, Camera Moves, Staging, Scene-to-Scene Transitions, Screen Aspect Ratios.

Soundtrack Recording and Editing: Talent Selection.

Storyboarding and Animatics: Storyboards, The Animatic.

UNIT-III

Digital Desktop Production: Stages of Animation Production, Production Team and Workflow, Project Management.

Principles of Animation: Key Poses, Breakdowns and Inbetweens, Timing, Extreme Positions, Arcs and Path Action, Holds, Emphasis, Anticipation, Weight and Weighted Movements, Overlapping Action, Generic Walks, Walk cycles, Runs and Run Cycles, Dialogue and Lip Sync, Laughter, Takes.

Animating Step by Step: Key Poses, Drawing for Animators.

UNIT-IV

2D Animation Overview: It's All About Pencil and Paper, Tools of the Trade.

2D Animation Basics: Keys, Inbetweens and Timing, Dope Sheets & Production Folders, Flipping & Peg Bars.

Finessing 2D Animation: Tracebacks, Eccentric Movement and Staggers, Panning & Camera Moves, Pan Speed & Strobing Problems, Shadows and Effects.

2D Vector Animation: Basic Approach, Vector Film Production, Being Resourceful, Non-Web Vector Animation.

The Paperless Animation Studio: The Technology, Importance of Drawing

References:

1. Tony White, 2006: *Animation: From Pencil to Pixels*, Focal Press.
2. Isaac V. Kerlow :The Art of 3D Computer Animation and Effects, John Wiley & Sons.
3. Mary Murphy, 2008 :Beginner's Guide to Animation: Everything you need to know to get started, Watson-Guptill Publications.
4. Yueting Zhuang, Yunhe Pan, Jun Xiao, 2008: *A Modern Approach to Intelligent animation: Theory and Practice*, Springer Publications

3ds Max: Modeling

Paper Code: MGA-204

Course Objective: introduce students to the principles and basic techniques of 3D Modeling .

UNIT I

Exploring the Interface: Interface Elements, Menus, Toolbars, Viewports, Command Panel, Lower Interface Bar Control, Workspaces.
Controlling & Configuring the Viewports: 3D Space, Navigation Gizmos, Using the Viewport Navigation Controls, Changing Viewport Display.
Working with Files: 3ds Max Scene Files, Importing and Exporting, File Utilities, Accessing File Information.

UNIT II

Setting Preferences: General, File, Gamma & Look-Up Table, Rendering & Radiosity, Animation, Kinametic, Gizmo, Mental Ray, Global Container Preferences.
Primitive Objects: Selecting System Units, Creating Primitive Objects, Exploring the Primitive Object Types, Architecture Primitives, Modeling Helpers.
Selecting Objects & Using Layers: Selecting Objects, Object Properties, Hiding & Freezing Objects, Using Layers, Scene Explorer, Object Properties, Display Properties.

UNIT III

Transforming Objects: Translating, Rotating & Scaling Objects, Transformation Tools, Pivot Points, Align Commands, Grids, Snap Options.
Cloning & Creating Object Arrays: Cloning Objects, Cloning Options, Mirroring Objects, Cloning Over Time, Creating Object Arrays.
Organizing Scenes: Containers, Referencing External Objects, File Link Manager, Schematic View Window, Hierarchies.

UNIT IV

Subobjects & Modifiers: Model Types, Normals, Subobjects, Modifiers, Modifiers Stack.
2D Splines & Shapes: Drawing in 2D, Editing in Splines.
Modelling with Polygons: Poly Objects, Creating Editable Poly Objects.
Graphite Modelling Tools, Freeform Tools, Selection Tools, Object Paint Tools.
Compound Objects: Types, Morphing Objects, Terrain Objects, Blob Mesh Objects, ProBoolean & Procutter Objects.

References:

1. Kelly L., Murdock, 2013 : *Autodesk 3Ds Max 2014 Bible*, Wiley.
2. Dariush Derakhshani, Randi L. Derakhshani, 2011: *Introducing 3ds Max 9-3D for Beginners,* Wiley India.
3. Kelly L. Murdock, 2013:*Mastering Autodesk 3ds Max 2013*, Wiley India.
4. Markus Kuhlo, 2013: *Architectural Rendering with 3ds Max and V-Ray*, Focal Press.
5. Todd Daniele, 2009: *Poly-modeling with 3ds Max: Thinking Outside of the Box, Focal Press.*

3ds Max: Texturing, Lighting & Rendering

Paper Code: MGA-205

Course Objective: introduce students to the principles and basic techniques of Texturing, Lighting & Rendering.

UNIT-I

Applying Textures: Understanding Material Properties, Working with State Material Editor, Using Shading Types, Using Specialized Material Types, Creating Compound Materials, Using Matt/Shadow Material.

Unwrapping UVS and Mapping Structure: Mapping Modifiers, Using the Unwrap UVW Modifier, Creating Baked Structure and Normal Maps.

UNIT-II

Working with Backgrounds Cameras and Lightening: Configuring and Aiming Cameras, Setting Camera Parameters, Camera Connection Modifier, Loading a Background Image and Camera.

Using Lights and Basic Lighting Techniques: Understanding the Basic of Lights, Types of Lights, Creating and Positioning Light Objects, Viewing a Scene from Light.

UNIT-III

Using Sunlight and Daylight System: Introduction of Sunlight and Daylight System, Understanding Azimuth and Altitude, Working with Light Tracing, Understanding Radiosity, Using Advanced Lighting Material.

Rendering:Working with Render Parameters, Adding Pre-Render and Post-Render Scripts, Rendering Stylized Scenes, Managing Render States.

UNIT-IV

Batch and Network Rendering: Batch Rendering Scenes, Understanding Network Rendering, Setting up a Network Render System, Setting up 3ds Max .

Rendering with Mental Ray and Iray: Enabling Mental Ray and Iray, working with Mental Ray and Iray, using Render Elements.

References:

1. Kelly L., Murdock, 2013 : *Autodesk 3Ds Max 2014 Bible*, Wiley.
2. Dariush Derakhshani, Randi L. Derakhshani, 2011: *Introducing 3ds Max 9-3D for Beginners,* Wiley India.
3. Kelly L. Murdock, 2013:*Mastering Autodesk 3ds Max 2013*, Wiley India.
4. Markus Kuhlo, 2013: *Architectural Rendering with 3ds Max and V-Ray*, Focal Press.
5. Todd Daniele, 2009: *Poly-modeling with 3ds Max: Thinking Outside of the Box, Focal Press.*

Practical Based on Paper MGA-204 & MGA-205

Paper Code: MGA-206

There will be a practical examination in this paper along with semester examination. External examiner will evaluate the skills of the students in the field of animation using 3ds Max. Examiner will give on-the-spot assignments to the students. Besides this a viva voce (oral examination) of each student one by one will also be taken by the examiner for testing the knowledge of the students.

Semester V

Paper Code: GFC-302

Internet Programming with Java

Course Objective: Understanding the core part of programming.

UNIT I

Java Script: Features, tokens, data types, variables, operations, control structs strings arrays,functions, core language objects, client side objects, event handling. Applications related to client sideform validation.

UNIT II

Fundamentals of Java: Java Vs. C++, Byte lode, Java virtual machine, constants, variables, data types, operators, expressions, control structures, defining class, creating objects, accessing class members, constructions, method overloading.

UNIT III

Inheritance : Basics, member access, using super to call super class constructors, creating a multi level hierarchy, method overriding, dynamic method dispatch, using abstract classes, using Final. Packages and Interfaces: Defining a package,Access protection:Importing packages, Interfaces, Defining an Interface, Implementing. Interfaces.

UNIT IV

Multi-threaded Programming: The Java Thread model, Thread priorities, Synchronizations,Messaging. The thread class and run able interface, The Main Thread : Creating a Thread,Implementing Run able, Extending Thread, Creating Multiple Threads, Thread Priorities;Synchronizations : Methods, Statements, Inter Thread Communication, Deadlock, Suspending,Resuming and Stopping Threads.

References:

1. Phillips LEE and Darmell: Computer Graphics, Second Edition, PHI, New Delhi.
2. Daniel Dang, 2010: An Introduction to Java Programming, PHI, New Delhi.
3. Balaguruswamy, E., 1998: Programming with Java, A Primer, TMH, New Delhi.
4. Wanger & Wyke, 2000 : Java Script Unleased, Techmedia, New Delhi, 2000.

3D Max Animation (Practical)

(Paper Code: MGA-301)

Course Objective: this subject enables students to grasp the basics of drawing, sketching, shading and (indoor and outdoor) modeling. Students gain experience in modeling, textures, composition, and design and world class animation.

Unit-1

Forest Scene in 3DS Max:

- Preparing the Scene
- Camera Setup
- Creating Light Sources
- Texture the Scene
- Fine Tuning
- Final Render Settings

Unit-2

Environment Modeling:

- Preparing the Scene
- Camera Setup
- Lighting & Texturing
- Final Render Settings

Unit-3

Bedroom, Kitchen, Bathroom Interior Design:

- Preparing the Scene
- Camera Setup
- Create Light Sources
- Texture the Scene
- Fine-Tuning
- Final Render Settings

Unit-4

House Exterior:

- *Preparing the Scene*
- Create Additional Materials
- Light Inside
- HDRI Illumination
- Fine-Tuning Shadows on the facade
- Final Render Settings

Text Books

1. Name of Authors Title of the Book Publisher
2. 3dsmax7 Fundamentals New Riders
3. TedBoardman 3d'sMax5Fundamentals Tech media
4. Inside 3ds max7 NewRiders
5. Michelebousquet Modelrig,Animate with 3d's max6 Many world production
6.

Reference Books

1. Michael E. Mortenson 3D Modeling, Animation, and Rendering Create space
2. Boris Kulagin 3ds Max 8 from Modeling to Animation Bpb
3. Michael G. 3D Modeling and Animation Igi Publishing

Visual Effects (Practical)

Paper Code: MGA-302

Course Objective: The subject aims at teaching the fundamentals of animation and visual effects, one of the key factors in the animation design works.

- Composition, visual aesthetics, technical proficiency, and artistic merit **50%**
- Completeness of project and clear communication of a unified concept **50%**

1. Self Portrait (Chroma Key) - (Photoshop, after effects) Using media generated in class and images and video created and collected from the first homework assignment, compose a brief (30 seconds) self portrait video. Use any stills, moving images, titles, effects, music or sound that you wish. Final product must include: sound, animated stills, applied effects, chroma key video, transitions, and titles.

1. **Title Sequence (as 3D motion graphic choreography) –** (after effects, Photoshop, illustrator)
   Create a title sequence for an invented television show or movie. Final version should be 30 seconds to 2 minutes long, utilize animated and/or processed text with layers of moving images. The final piece may be abstract or representational in its presentation. The project should emphasize 3D space, composition, design, and type in motion**.**

4. **Film Festival Show Opener –** (after effects, final cut, photoshop, etc…)
   Create a concept for an opening sequence to a film festival (either the EMF Media Arts Festival or another festival of your choice). The opener should be 30 seconds-1 minute in length and is open ended in terms of content, but it must utilize digital tools and involve 3D compositing, lighting, and camera movement in After Effects.

4. DVD and Demo Reel – (DVD studio Pro and any other software as needed) Compile all of your video work from the class onto DVD. The interface of the DVD must be original (NO templates), with menus, Transitions, and navigation all from scratch. Along with each project compiled on the DVD you must also edit together a short 30-second reel showcasing the highlights of your motion graphic work from the semester.
The 30-second highlight reel should be included on the compilation DVD and also compressed and uploaded to either google video, youtube, vimeo, or another similar site.

Design a DVD cover and disc label. The graduate students are expected to also include work from outside of the class in the Demo Reel and must utilize animated transitions/effects for DVD menus and button transitions.

PRE PRODUCTION (THEORY)
Paper Code: MGA-303

Course Description:

Pre-production refers to the tasks that must be completed or executed before filming or shooting begins. This includes tasks such as hiring actors or models, building sets, budgeting, planning, scheduling, renting equipment and tests, to name a few of the many pre-production tasks.

Course Goals:
The goal of pre-production is to develop an efficient structure for your project on which the final animation will be produced. In pre-production, we identify potential difficulties in production and work with you to minimize or eliminate them.

Course Activities and Schedule:

UNIT: 1
Development backbone –Identifying needs and research -Visualize your project‘s Look -Collect images that inspire you visually

UNIT: 2
Review Process and Pre-Production Schedule -Constant communication
-Content decisions incrementally throughout all stages of production
-Maximum efficiency and identify unforeseen and time-consuming changes.

UNIT: 3
Find and Secure Location-Available light in the location -Size of the location –Cause of sound issues – Casting -Make sure your posting / flyer contains relevant info about the roles you are casting (include age, physical characteristics) short blurb describing character, date /time / length of audition, materials that need to be prepared (such as monologue), your contact info, shoot timeframe, remuneration -Local casting resources -Videotape auditions if possible.

UNIT: 4
Production Design -Film‘s theme -Mood progressions -Kind of location should each sequence have -Color palette and progression that promote the film‘s thematic development -Prepare costumes, props, set dressing.

UNIT: 5
Make a Floor plan -Lighting plot for each Location –Outline and Scripts -Break down, shooting script - Accurate and effective script that engages your audience -Make a Storyboard -Composition of the scenes, the position of the camera, the story‘s theme and characters.

Suggested books for Reading:

- The Five C's of Cinematography: Motion Picture Filming Techniques , by Joseph V. Mascelli
- Single Camera Video Production, by Robert B Musburger.

Semester VI

Project & Internship.

Paper Code: MGA-304

Rationale

The main idea behind internship/training/apprenticeship is to provide an opportunity to the students to have hands-on-experience being a team member of a graphics/animation/gaming project in a real life industry environment so that s/he could achieve a minimum level of confidence in production skills required in the beginning of a career.

Introduction

Each student shall be supposed to complete an internship/training /apprenticeship during the last semester of the course. This will be arranged by the institute or student in consultation with the teacher concerned.

Process

Each student will complete his/her six months internship/training/ apprenticeship in a national or international company/institution/ any other organization which is actively engaged in the production of graphics, animation and gaming. The final approval regarding the selected companies/institute/organization will be given by the Head of the Institute.

Evaluation

During the internship/training/apprenticeship students will work under a supervisor to be decided by the production house. In the end of the training supervisor will evaluate the student on the basis of followings:

- Punctuality in completing assignments
- Participation in project work
- General behaviour during training
- Ability to work as a team member
- Learning abilities and work performance

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